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뇌-마음-행동 개별연구 보고서

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연구 제목	Cognitive Mechanisms of Non-invasive Brain Stimulation for Substance Use Disorders and Behavioral Addictions: A Systematic Review

1. 연구의 배경 및 중요성

Across a large number of studies on addiction, it has been documented that the disorder fundamentally involves cognitive impairments, namely maladaptive changes in motivational processes such as reward and risk processing, as well as weakened top-down regulation over these functions, ultimately leading to impaired decision making over addictive use/behavior and poor treatment outcomes (see Ramey & Regier, 2019, for review). Hence, for significant and long-lasting effects of addiction treatments, normalizing the modified cognitive functions in addicted individuals is a crucial step in the therapeutic journey.

Previous literature has demonstrated that non-invasive brain stimulation (NIBS) techniques, including transcranial magnetic stimulation (TMS) and transcranial direct current stimulation (tDCS), are effective in treating clinical symptoms of addiction, such as craving or consumption. Although somewhat promising, several reviews on this topic have consistently reported their limitations, with interventions yielding small effects and variant results across different study samples and stimulation protocols (Mehta et al., 2024; Naish et al., 2018; Stein et al., 2019). With that said, these studies have pointed out the importance of investigating the alterations in neurocognitive processes that underlie these effects, in order to investigate the differing neural and cognitive states that may reinforce addictive behavior across individuals in distinct manners.

Recent advances in neuromodulation techniques have sparked an increasing interest in targeting specific neural circuits underlying these cognitive processes for treating substance use disorders and behavioral addictions, with risk-taking, impulsivity, attention, and cognitive control as common domains of improvement (Trojak et al., 2017). Numerous studies on NIBS have supported its capability to induce cognitive changes in addicted individuals, enhancing decision making and control undermined due to addiction, and debilitating excessive automatic responses towards substance or addictive behavior (see Naish et al., 2018, for review).

While NIBS effects on addictive symptoms may be inconsistent, investigating the cognitive improvements it generates may better support the efficacy of this treatment method, capturing subthreshold changes in the underlying cognitive processes which clinical measures do not fully incorporate. Furthermore, a comprehensive examination of this topic is crucial for a significant development in the clinical treatment of addiction, as applying these findings in tailoring neurostimulation methods to maximize treatment effects for individuals with distinct cognitive profiles should lead to a more effective, reliable, and durable treatment.

In this vein, the current study examines where we stand in understanding the latent cognitive targets for brain stimulation in addiction treatment. Reviewing the available research on NIBS interventions directed towards cognitive impairments in clinical and subclinical populations with addiction, this article aims to present the efficacy and limitations of NIBS treatments in modifying cognitive processes relevant to addiction. Specifically, NIBS effects on alterations in several cognitive domains including decision-making, bias, regulation of automatic responses, and reward updating, are discussed in detail. This comprehensive review of NIBS treatment on addiction is significant both as a thorough summary of up-to-date findings in this growing research area, and as a potential bridge between research and the clinical applications of NIBS-based addiction treatment.

2. 연구 방법 및 결과

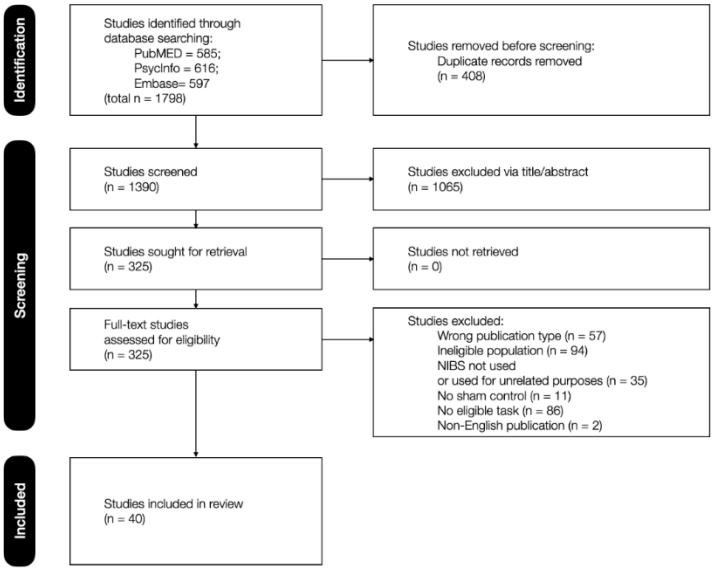

Figure 1. Flow diagram of screening process for the systematic review. The search yielded 1798 articles across PubMed, PsycInfo, and EMBASE, and eliminating duplicates left 1398 articles to be screened. After the title and abstract screening, 325 articles went through full-text assessment to check for appropriate study population and NIBS-intervention methodology. Finally, 40 articles were determined as eligible for the current review.

This systematic review followed the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) guidelines (Page et al., 2021). Studies for review were searched on PubMed, PsycInfo,

and EMBASE, covering studies published between 2000 and 2025. The detailed query used for search is indicated in Appendix 1, grouped as addiction-related, NIBS-related, and cognition-related keywords. The entire screening process is thoroughly summarized in **Figure 1.**

The articles included in this review addressed distinct cognitive domains closely related to addiction, which we group as the following four categories: attentional and behavioral bias, response inhibition and inhibitory control, valuation and decision-making, and reward processing and learning.

A larger number of articles were found in the first two categories, which involved implicit, automatic, and relatively low-level cognitive processes compared to the latter two. To begin, 10 articles targeted attentional and behavioral bias towards addiction-related stimuli, mostly using the Implicit Association Test (IAT) and the Approach-Avoidance Task (AAT) to capture and modulate the implicit positive associations with addictive stimuli and biased approach tendency towards substance-related cues in individuals with addiction **(Table 1)**. These studies mostly demonstrated null effects of tDCS in improving implicit associations relevant to addiction represented in the IAT, as well as inconsistent efficacy in modulation of automatic approach behavior towards addictive stimuli. Furthermore, although neuromodulation techniques were often paired with cognitive bias modification (CBM) techniques for enhanced treatment outcomes, the synergistic effects were inconsistent across these studies. The lack of significant results can be partially attributed to the shared characteristics of the recruited samples in a subset of these studies (den Uyl et al., 2017, 2018; Schwippel et al., 2022, 2025), which recruited either recently abstinent and/or treatment-engaged participants. These studies revealed that participants showed baseline avoidance biases instead of the expected approach bias, leaving small room for NIBS-induced improvement. Such observation suggests the importance of a well-timed intervention within the temporal course of an addiction cycle in order to retrieve meaningful treatment effects. Notably, explicit ratings of stimuli and IAT bias scores showed consistent positive correlations, suggesting the dynamic interactions and mutual influence of top-down and bottom-up operations relevant to improvements in addictive behavior. However, it seems that interventions targeting cognitive bias in addiction have yet to fully incorporate such findings into sophisticating their methodologies.

The largest number of studies in our review targeted response inhibition or inhibitory control, which are widely accepted as a central executive function impaired in addictive populations, contributing to incessant engagement in addictive behavior. 20 studies were categorized under this group, most of them using the Stop Signal Task (SST) or Go/No-Go task (GNG) to measure ability to withhold or stop an automatic response **(Table 2)**. Results from these studies demonstrated that NIBS targeting the dlPFC can enhance response inhibition in individuals with addiction, although the large variance in findings across tasks, sample characteristics, and protocols, cannot be ignored. In general, GNG yielded more consistent NIBS effects compared to SST, which may be due to the dissimilar nature of these tasks. While the former captures the (in)ability to restrain the initiation of an action (i.e., proactive inhibition), the latter measures the ability to cancel an initiated response (i.e., reactive inhibition). In this sense, GNG may rely more on prefrontal processes directly modulated by NIBS, while SST involves motor-level inhibition based on subcortical mechanisms, resulting in different outcomes across task types. In support of this, neuroimaging and electrophysiological methods coupled with NIBS targeting prefrontal regions have revealed enhanced prefrontal-cingulate connectivity, representing strengthened top-down regulation over implicit processes following the interventions. Hence, NIBS-enhanced prefrontal control circuits seem to be relatively effective in promoting response inhibition compared to the more low-order regulatory system represented in the SST.

The remaining two groups of articles touched on more complex and elaborated cognitive processes, namely valuation and decision-making, as well as learning and habitual behavior. A total of 14 articles aimed to alleviate deficits in risky or impulsive decision-making in addictive populations, a central mechanism related to the recurrent use of substance and maintenance of addictions **(Table 3)**. The tasks used in these studies included the Balloon Analogue Risk Task (BART) and the Game of Dice Task (GDT), measuring risk-taking levels under conditions with known or unknown probabilities of risk, and delay discounting tasks (DDT), used for quantifying the degree to which individuals undermine the values of delayed rewards and make impulsive choices. The studies delivered promising results, with reduced risk-taking levels found even in single-session NIBS treatments, although repeated sessions generated more consistent effects. Regarding

delay discounting and impulsivity, the heterogeneity of results was evident, with improvement in these cognitive domains emerging only occasionally among other studies illustrating no changes in task performance following NIBS treatment. Evidence points to the superior effects of NIBS combined with cognitive or behavioral training interventions, where stimulation-induced neuroplasticity interacts with and facilitates cognitive improvement triggered by such training. Although the neural mechanisms responsible for the changes described above are still to be investigated, alterations in prefrontal-striatal circuitry and improved GABA (inhibitory) and glutamate (excitatory) neurotransmitter balance are possible candidates supported by existing evidence.

Last but not least, 8 articles explored the possible effects of NIBS on modulating dysfunctional reward processing and learning patterns in individuals with addiction, which are core cognitive components integral to the nature of addiction by which individuals fail to incorporate the harmful outcomes of addictive behavior into their future actions **(Table 4)**. These studies mainly used the Monetary Incentive Delay (MID) task, and the T-maze task, in order to target blunted sensitivity to rewards in addictive populations, and Iowa Gambling Task (IGT), to capture maladaptive learning processes under ambiguity in these individuals. Findings illustrate the capability of prefrontal rTMS in enhancing reward-related neurophysiological responses, specifically an ERP signal named "reward positivity." Reward positivity reflects the neural reaction towards the difference between expected and gained rewards, or reward prediction error (RPE), an essential signal for value updating in reinforcement learning. However, NIBS-effects greatly varied among behavioral measures of reward-based learning across task types, with virtually no changes found in reward anticipation measured in MID while enhanced task performance was consistently found across different populations and NIBS techniques in IGT. Yet, it should be noted that the latter findings were constrained to reporting solely the overall augmentation in IGT performance (e.g., higher net scores). Considering that IGT is a highly sophisticated task encompassing learning, ambiguity processing, and decision making, a refined framework integrating computational parameters to dissect the diverse and interrelated components of this task is needed.

3. 연구의 의미

In summary, this review highlighted the importance of targeting specific cognitive mechanisms instead of aiming for global effects on clinical symptoms in NIBS-based addiction treatment. Here, documented attempts to mitigate cognitive dysfunctions with the purpose of treating addiction are reported under four distinct categories of cognition: bias, inhibition, decision-making, and reward-based learning. Although results have been variable, studies have illustrated significant NIBS-induced improvements in maladaptive cognitive functions characteristic of addictive populations.

While former studies have tilted towards targeting more implicit, rudimentary areas of cognition, future studies should make more use of intricate cognitive tasks, integrating the complex and dynamic processes of decision making and learning relevant to addiction. Importantly, findings from these tasks should not be summarized into an abstract, ambiguous cognitive function, but rather be deconstructed into specific, interrelated components, enabling an elaborated assessment of cognitive dysfunction contributing to continuous addictive behavior. Among the studies screened for review, Liu et al. (2024) had used a computational modeling approach to map IGT results onto defined parameters indicating learning rate, outcome sensitivity, loss aversion, etc., fully employing the complicated design of the task to analyze various elements of decision-making and learning. In this vein, future studies should adopt computational models for a more refined understanding of addictive behavior and identification of possible intervention targets. Specifically, the Reinforcement Learning and Decision-Making (RLDM) framework, which models specific steps within learning and decision-making processes (i.e., valuation, action selection, outcome evaluation, and learning), holds great potential as a powerful tool to be applied in the addiction treatment setting. By providing relevant tasks and precise language to illustrate the altered cognitive processes across different stages of addiction, such a computational framework would allow for a more holistic, sophisticated, and individualized understanding of the impaired cognition in addictive populations.

Table 1. Summary of Studies on Attentional or Behavioral Bias (n=10)

Study	Sample	InterventionProtocol			Outcome Measures (Addiction / Cognitive / Other)	Key Findings	
		Method	Target	Parameters		Clinical	Behavioral/ Cognitive
Alcohol							
den Uyl et al. (2015)	Hazardous drinkers Alcohol use measured by AUDIT (>8)	tDCS	Left DLPFC F3 anode / Right IFG crossing of F7, CZ, Fz and T3 anode / Contralateral supraorbital region cathode	1 mA 10 min Single session	Addiction: Craving (AAAQ), Alcohol use; AUDIT, Retrospective 1 week alcohol diary Cognitive: IAT	DLPFC stimulation reduced craving compared to sham IFG stimulation showed no significant reduction	- DLPFC stimulation improved reaction times for attribute words in IAT - No effects on IAT bias scores (difference between alcohol-positive and alcohol-negative trials)
Vanderhasselt et al., 2020	Heavy drinkers Alcohol use measured by AUDIT (>16)	tDCS	Bilateral DLPFC F4 anode F3 cathode	2 mA 20 min Single session	Addiction: Alcohol consumption (Beer-tasting test), Craving (AAAQ) Cognitive: Rewarded Go/No-Go (reward triggered approach bias)	Reduced alcohol consumption after active tDCS	Reduced reward-triggered approach bias after active tDCS
Claus et al. (2019)	At-risk drinkers Alcohol use measured by AUDIT (>8)	tDCS, CBM (Motivaitonal Bias Retraining)	Right IFG F10 anode Contralateral upper arm cathode	2 mA 20 min 4 sessions	Addiction: Alcohol drinking quantity/frequency (drinks per drinking day, % heavy drinking days) Cognitive: AAT	- No significant tDCS effects on alcohol drinking quantity/frequency - Trend for decrease in drinks per drinking day over time	No significant tDCS effects on approach bias
Schwippel et al. (2025)	AUD ICD-10 criteria Recently abstinent and participated in therapeutic interventions	tDCS	Left DLPFC Right deltoid muscle anode F3 cathode	2 mA 30 min Single session	Addiction: Craving (Numeric Analog Scale) Cognitive: IAT & Explicit alcohol-related cognition (alcohol approach, drinking identity)	Increased nicotine craving during experimental session in sham tDCS group but not active	- No tDCS effects on cognitive tasks - IAT and explicit ratings indicated alcohol-avoidance and non-drinking identity bias - D-IAT scores and explicit ratings were positively correlated
Schippel et al. (2022)	AUD ICD-10 criteria Recently abstinent patients	tDCS	Left DLPFC Right deltoid muscle anode F3 cathode	1 mA 30 min Single session	Addiction: Craving (VAS) Cognitive: IAT & Explicit alcohol-related cognition (alcohol approach, drinking identity)	No significant tDCS effects on craving	- No tDCS effects on cognitive tasks IAT scores indicated alcohol-avoidance and non-drinking identity bias - IAT effects and explicit ratings were positively correlated
den Uyl et al. (2017)	Alcohol dependent inpatients Received inpatient treatment during study	tDCS, CBM (Alcohol approach bias retraining)	Bilateral DLPFC F3 anode F4 cathode (big electrode)	2 mA 20 min 4 sessions	Addiction: Craving (PACS), Relapse (3 month, 1 year follow-up) Cognition: AAT, IAT, Stroop task	- No intervention effect on relapse measured after 3 months - Trend level significant decrease in relapse measured after 1 year in active tDCS+CBM compared to sham tDCS+CBM	- No intervention effect across all sessions - During training, active tDCS+CBM showed bigger decrease from session 1 to session 2 compared to sham tDCS+CBM
den Uyl et al. (2018)	Alcohol dependent inpatients Received inpatient treatment during study	tDCS, CBM (Attentional Bias Modification)	Bilateral DLPFC F3 anode F4 cathode (big electrode)	2 mA 20 min 4 sessions	Addiction: Craving (PACS),, Relapse (1 year follow-up) Cognitive: Alcohol Visual Probe Task, Alcohol approach/avoidance IAT	- No intervention effects on craving or relapse - The least amount of relapse found in patients from active tDCS groups	- No intervention effect across all sessions - During training, negative alcohol bias was only found in active tDCS+real ABM
den Uyl et al. (2016)	Hazardous drinkers Alcohol use measured with AUDIT (>8)	tDCS, CBM (Alcohol approach bias retraining)	DLPFC F3 anode	1 mA 15 min 3 sessions	Addiction: Craving (AAAQ), Alcohol use, Cue-reactivity, Readiness to Change (RCQ) Cognitive: AAT, IAT, Alcohol Oddball Task	- Decreased cue-induced craving for alcohol after active tDCS compared to sham tDCS. - No significant intervention effect on overall craving.	- No CBM effects on alcohol bias - No enhancement effects of tDCS on CBM - During training, participants in active CBM compared sham showed alcohol avoidance bias in session 1
Nicotine							
Mondino et al. (2020)	Tobacco use disorder DSM-5 criteria	tACS, ABM	Bilateral DLPFC F3 F4	2 mA 10 Hz (alpha) 30 min Singles session	Addiction: Craving (Questionnaire of smoking urges) Addiction: Attentional bias (Passive observation task, Visual Probe Task), Delay Discounting Task*	Reduced self-reported desire to smoke after active tDCS+ABM compared to sham tDCS+ABM	- Reduction in time spent looking at smoking-related pictures after active tDCS+ABM compared to sham tDCS+ABM - No intervention effects on VPT
Methamphetamine							
Xu et al. (2021)	Methamphetamine use disorder DSM-5 criteria Abstinent for 13-16 months	tDCS	Bilateral DLPFC F4 anode F3 cathode	1.5 mA 20 min 20 sessions	Addiction: Cue-induced craving (VAS) Cognitive: Dot Probe Task, SST*, Delay Discounting Task*	Only CCAT with active tDCS reduced cue-induced craving after the 4-week intervention	- Delay discounting improved over time in intervention groups, while control group performed worse over time (protective effects of CCAT+active/sham tDCS) - No significant change in SST

Table 3. Summary of Studies on Response Inhibition and Inhibitory control (n=20)

Study	Sample	InterventionProtocol			Outcome Measures (Addiction / Cognitive / Other)	Key Findings	
	Population	Method	Target	Parameters		Clinical	Behavioral/Cognitive

Alcohol							
Holla et al. (2020)	AUD DSM-5 criteria	tDCS	Bilateral DLPFC F4 anode F3 cathode	2 mA 20 min 5 sessions	Addiction: Craving (ACQ-SF-R) Cognitive: SSRT, Go/No-Go Other: rsfMRI	- No significant tDCS effect on craving 0 Longer time to first alcohol use in active condition - Increased global efficiency predicted reduced likelihood of relapse	Trend level reduction in impulsivity (SSRT)
del Felice et al. (2016)	AUD DSM-4 criteria	HF-rTMS	Left DLPFC F3	10 Hz 100% MT 4 sessions	Addiction: Craving (VAS), self-reported alcohol intake Cognitive: Stroop test, Go/No-Go Other: Resting EEG	- Both groups showed reduction in depressive symptoms and somatization -Somatization/depression in active group was significantly correlated with theta/alpha bands over central regions	- Go/No-Go performance increased over time - Stroop test was inversely correlated with faster frequencies (alpha2 and beta bands)
Dormal et al. (2020)	Binge drinkers Binge drinking measured by ???	tDCS	Left DLPFC F3 anode F4 cathode	1.5 mA 20 min 2 sessions	Cognitive: Go/No-Go (neutral, alcohol-related), UPPS-P Other: EEG/ERP (N2, P3 components)	Binge drinking score was negatively correlated with correct No-Go responses in BD	- No significant behavioral/cognitive changes - Active tDCS increased N2 amplitude in BD for correct Go trials, and P3 amplitude in non-BD for correct No-Go trials - Both groups showed reduced N2 latency for Go trials UPPS-P score was correlated with correct No-GO responses in sham-tDCS
Dubuson et al. (2021)	Severe AUD Require rehabilitation	tDCS, Alcohol cue inhibitory control training	Bilateral DLPFC F4 anode F3 cathode	2 mA 20 min 5 sessions	Addiction: Craving (VAS), Alcohol consumption (Timeline flowback) Cognitive: SSRT	- Active tDCS showed 19% higher abstinence rate at 2 weeks compared to sham - Active tDCS-AICT showed highest abstinence compared to other conditions - Effects on abstinence did not persist beyond two weeks No significant changes in craving	No significant changes
Weidler et al. (2022)	Alcohol dependent patients (ICD-10 criteria) Tobacco users	tDCS	Right DLPFC F4 anode Contralateral supraorbital area cathode	1.5 mA 20 min Single session	Cognitive: SSRT Other: Aggression (mTAP)	N/A	- No post-stimulation increases in punishment selections (aggression) in only AD with active tDCS - Improved SSRT in AD and TU following active tDCS but not sham tDCS.
da Silva et al. (2013)	Lesch's Type IV alcohol dependent patients (premorbid cerebral defects) DSM-4 criteria	tDCS	Left DLPFC F3 anode Right supradeltoid area cathode	2 mA 20 min 5 sessions	Addiction: Craving (OCDS), Relapse (4 week), Alcohol Urge Questionaire Cognitive: Go/No-Go Other: EEG/ERP	Significant reduction in depression and craving after active tDCS compared to sham High relapse rate in tDCS compared to sham	Trend for improved executive function in tDCS
Daughters et al. (2020)	Substance use disorder (alcohol, cocaine, opiate, cannabis) DSM-5 criteria	tACS	Bilateral DLPFCS F3 F4 Cz reference electrode	2 mA peak-to-peak Alpha-tACS: 10 Hz Gamma-tACS: 40 Hz 2 sessions	Cognitive: Go/No-Go, Distress tolerance (PASAT-C)* -	N/A	- Large and significant effect of alpha-tACS compared to sham-tACS - No significant tACS effects in distress tolerance
Schluter et al. (2019)	AUD DSM-5 criteria Had intention to quit	HF-rTMS	Right DLPFC F4	10 Hz 110% MT 10 sessions	Cognitive: Go/No-Go, SST, Delay Discounting Task* -	N/A	- No significant change - No significant correlations between task performance and baseline impulsivity/severity of AUD - Higher age predicted better GNG performance
Nicotine							
Müller et al. (2021)	Tobacco use disorder (TUD) DSM-5 criteria Non treatment seeking	tDCS	Bilateral DLPFC F3 anode F4 cathode	2 mA 20 min 5 sessions	Addiction: Cigarette consumption,, Craving (VAS) Cognitive: Go/No-Go, SST, Delay Discounting Task* -	- Decreased cigarette consumption, CO levels, and post-stimulation craving in both groups - Higher perceived stress predicted lower cigarette consumption	- No cognitive/behavioral improvement after stimulation - Lower self-control predicted higher CO reduction
Song Y et al. (2025)	Internet Gaming Disorder (DSM-5, ≥6 criteria)	tDCS (cue-synchronized)	Central Parietal (Pz cathode)	1.5 mA 20 min 2 sessions	Addiction: Craving (QGU-B, VAS) Cognitive: SST, MID task, IGT* Other: EEG (P300, alpha) during cue reactivity task	Craving and gaming time decreased on active tDCS group (QGU-B: p<.001, weeks 1-3; VAS: p<.001, weeks 2-3; gaming time: p<.001, weeks 3-4)	- No significant change in MID (d' or reaction time) - No significant tDCS effect on SST - IGT net score increased after active tDCS compared to sham at 1 week, 4 week follow-up. SST: no significant group x time interaction for all measures
Internet Gaming							
Jeong et al. (2024)	Internet Gaming Disorder DSM-5 criteria	tDCS	Bilateral DLPFC F3 anode F4 cathode	2 MA 30 min 10 sessions	Addiction: Craving (VAS) Cognitive: SSRT Other: rsfMRI	Craving decreased over time	- Active group showed decreased SSRT (improved inhibition) - ACC-left MFG connectivity negatively correlated with SSRT in active group

Wu et al. (2021)	Internet gaming disorder DSM-5 criteria	tDCS	F4 anode Left rapezius muscle cathode	1.5 mA 20 min 2 sessions	Craving (background, cue-induced) Letter categorization task with gaming-related distractors -	- Active tDCS reduced background craving measured on a 9-point scale - No significant change in cue-induced craving	- Active tDCS reduced interference from gaming compared to non-gaming pictures
Gambling							
Gavazzi et al. (2025)	Gambling disorder DSM-5 criteria	cTBS	pre-SMA	5 Hz 80% of active MT 1200 pulses Single session	Cognitive; SST	N/A	- Active TMS group showed decreased SSRT and increased SSD compared to pre-treatment and non-treated HCs
Zack et al. (2016)	Pathological Gambling DSM-4 criteria Non-treatment seeking	HF-rTMS, cTBS	rTMS: mPFC cTBS: rDLPFC sham: Vertex	rTMS: 10 z, 80% MT cTBS: 5 Hz, 80% MT Single session	Addiction: Desire to gamble (VAS), Psychostimulant-like sensations (ARCI), Physiological arousal (Blood pressure response to game) Cognitive; Commercial slot game, Stroop task, Delay Discounting Task	- Less increase in desire to gamble after active rTMS compared to sham - Reduced amphetamine-like effects and decreased diastolic blood pressure after cTBS compared to sham	- Reduced behavioral activation during game after treatment compared to sham - Worsened attentional control (Stroop) after both treatments - No stimulation effect on DDT
Methamphetamine							
Xu et al. (2021)	Methamphetamine use disorder DSM-5 criteria	tDCS	Bilateral DLPFC F4 anode F3 cathode	1.5 mA 20 min 20 sessions	Addiction: Cue-induced craving (VAS) Cognitive: SST, Dot Probe Task*, Delay Discounting Task*	Only CCAT with active tDCS reduced cue-induced craving after the 4-week intervention	- Delay discounting improved over time in intervention groups, while control group performed worse over time (protective effects of CCAT+active/sham tDCS) - No significant change in SST
Alizadehgoradel et al. (2021)	Methamphetamine use disorder DSM-4 criteria Early abstinent users	tDCS	Bilateral DLPFC F3 anode F4 cathode	1.5 mA 20 min 12 sessions	Addiction: Craving (DDQ) Cognitive: Go/No-Go, BART*	Significant craving reduction compared in all active treatment groups compared to sham, with strongest effects maintained in tDCS+MBSAT group at 1-month follow-up	- Combined tDCS+MBSAT showed superior improvements in BART and GNG performance compared to other groups - Significant correlations between craving reduction and improvements in BART performance in tDCS+MBSAT group
Alizadehgoradel et al. (2020)	Methampehtamine use disorder DSM-5 criteria	tDCS	Bilateral DLPFC F3 anode F4 cathode	2 mA 20 min 10 sessions	Addiction: Craving (DDQ) Cognitive: BART, Go/No-Go*	Craving reduction in active-tDCS group compared to baseline and sham-tDCS immediately after stimulation and at 1 month follow-up	- Improved GNG, BART performance in active-tDCS group compared to baseline and sham-tDCS, immediately after stimulation and at 1 month follow-up - Significant correlation between task performance improvement and craving reduction
Opioid							
Aksu et al. (2024)	Opioid use disorder DSM-5 criteria	tDCS, Cognitive Training (CT)	Bilateral DLPFC Right anode Left cathode	2 mA 20 min Single session	Cognitive: SST, Stroop, IGT*, BART*	N/A	Improved decision making under ambiguity (IGT) after active tDCS compared to sham
Borzooee et al. (2024)	Opioid use disorder under treatment DSM-5 criteria	tDCS	Right DLPFC F4 anode FP1 cathode	2 mA 13 min 10 sessions	Addiction: Instant craving (DDQ), Periodic drug craving (OCDUS), Cue-induced craving (VAS) Cognitive: BART, Go/No-Go* Other: Serum BDNF	- Periodic drug craving (OCDUS) sigfniciantly decreased in active tDCS at 2 month follow-up - No significant intervention effects on instant (DDQ) or cue-induced craving (VAS)	- Risk taking (BART) decreased in active tDCS group after stimulation and at 2 month follow-up. - Response inhibition (GNG) decreased in active group after stimulation and at 2 month follow-up. - The above effects were not significant in the gerenalized eestimating model.
Cocaine							
Verveer et al. (2020)	Cocaine use disorder DSM-5 crieria	tDCS	Bilateral DLPFC Right anode Left cathode	2 mA 13 min 10 sessions	Addiction: Craving (Ecological Momentary Assessments), Relapse (3 month follow-up) Cognitive: Go/No-Go, 2-Choice Gambling Task*	No significant tDCS effects on craving nor relapse Reduction in relapse rates for crack cocaine use after active tDCS compared to sham	No significant tDCS effects on cognitive/behavioral measures

Table 3. Summary of Studies on Valuation and Decision-making (n=14)

Study	Sample	Intervention Protocol			Outcome Measures (Addiction / Cognitive / Other)	Key Findings	
		Method	Target	Parameters		Clinical	Behavioral/ Cognitive
Alcohol							
Schluter et al. (2019)	AUD with intention to quit DSM-5 criteria	HF-rTMS	Right DLPFC F4	10 Hz 110% MT 10 sessions	Cognitive: Delay Discounting Task, Go/No-Go*, SST*	N/A	- No significant TMS effects on cognitive task performance
Nicotine							

Müller et al. (2021)	Non-treatment seeking tobacco use disorder (TUD) DSM-5 criteria	tDCS	Bilateral DLPFC F3 anode F4 cathode	2 mA 20 min 5 sessions	Addiction: Cigarette consumption,, Craving (VAS) Cognitive: Delay Discounting Task, Go/No-Go*, SST*	- Decreased cigarette consumption, CO levels, and post-stimulation craving in both groups - Higher perceived stress predicted lower cigarette consumption	- No cognitive/behavioral improvement after stimulation - Lower self-control predicted higher CO reduction
Lechner et al. (2022)	Non-treatment seeking, moderately dependent smokers Nicotine dependence measured by FTCD (≥ 5)	rTMS, Working Memory Training (WMT)	Left DLPFC F3	10 Hz 100% MT 10 sessions	Addiction: Smoking urge (VAS), Minnesota Nicotine Withdrawal Scale Cognitive: Delay to Smoking Analog Task, Cigarette purchase task,	No significant effect of intervention	Combining rTMS and WMT worsened performance on delay to smoking analog task
Mondino et al. (2020)	Tobacco use disorder DSM-5 criteria	tACS, ABM	Bilateral DLPFC F3 F4	2 mA 10 Hz (alpha) 30 min Singles session	Addiction: Craving (Questionnaire of smoking urges) Cognitive: Delay Discounting Task, Attentional bias* (Passive observation task, Visual Probe Task)	Reduced self-reported desire to smoke after active tDCS+ABM compared to sham tDCS+ABM	- Reduction in time spent looking at smoking-related pictures after active tDCS+ABM compared to sham tDCS+ABM - No intervention effects on VPT
Methamphetamine							
Xu et al. (2021)	Methamphetamine use disorder DSM-5 criteria Abstinent for 13-16 months	tDCS	Bilateral DLPFC F4 anode F3 cathode	1.5 mA 20 min 20 sessions	Addiction: Cue-induced craving (VAS) Cognitive: Delay Discounting Task, Dot Probe Task*, SST*	Reduced cue-induced craving after the 4-week intervention in CCAT+tDCS group	- Delay discounting improved over time in intervention groups, while control group performed worse over time (protective effects of CCAT+active/sham tDCS) - No significant change in SST
Alizadehgoradel et al. (2021)	Early abstinent methamphetamine users DSM-4 criteria	tDCS, MBSAT	Bilateral DLPFC F3 anode F4 cathode	1.5 mA 20 min 12 sessions	Addiction: Craving (DDQ) Cognitive; BART, Go/No-Go*	Significant craving reduction compared in all active treatment groups compared to sham, with strongest effects maintained in tDCS+MBSAT group at 1-month follow-up	- Combined tDCS+MBSAT showed superior improvements in BART and GNG performance compared to other groups - Significant correlations between craving reduction and improvements in BART performance in tDCS+MBSAT group
Alizadehgoradel et al. (2020)	Methamphetamine use disorder DSM-5 criteria	tDCS	Bilateral DLPFC F3 anode F4 cathode	2 mA 20 min 10 sessions	Addiction: Craving (DDQ) Cognition: BART, Go/No-Go*	Craving reduction in active-tDCS group compared to baseline and sham-tDCS immediately after stimulation and at 1 month follow-up	- Improved GNG, BART performance in active-tDCS group compared to baseline and sham-tDCS, immediately after stimulation and at 1 month follow-up - Significant correlation between task performance improvement and craving reduction
Gambling							
Dickler et al. (2018)	Gambling disorder DSM-5 criteria	tDCS	Bilateral DLPFC F4 anode F3 cathode	1 mA 30 min 2 sessions	Addiction: Craving (VAS) Cognitive: BART Other: MRS metabolite measurements (GABA, Glx, NAA)	- Craving was positively correlated with striatal Glx during active tDCS only	– Only during active tDCS: BART risk-taking was positively correlated with prefrontal Glx and striatal GABA during only active tDCS.
Zack et al. (2016)	Pathological Gambling DSM-4 criteria Non-treatment seeking	HF-rTMS, cTBS	rTMS: mPFC cTBS: rDLPFC sham: Vertex	rTMS: 10 z, 80% MT cTBS: 5 Hz, 80% MT Single session	Addiction: Desire to gamble (VAS), Psychostimulant-like sensations (ARCI), Physiological arousal (Blood pressure response to game) Cognitive: Commercial slot game. Stroop task*, Delay Discounting Task	- Less increase in desire to gamble after active rTMS compared to sham - Reduced amphetamine-like effects and decreased diastolic blood pressure after cTBS compared to sham	- Reduced behavioral activation during game after treatment compared to sham - Worsened attentional control (Stroop) after both treatments - No stimulation effect on DDT
Cocaine							
Gaudreault et al. (2021)	Cocaine use disorder DSM-5 criteria Detoxified for at least 2 weeks	tDCS	Bilateral DLPFC F4 anode F3 cathode	2 mA 20 min 15 sessions	Addiction: Craving (OCCS, Cocaine Selective Severity Assessment), Abstinence, Willingness to change drug use (contemplation ladder), STRAP-R Cognitive: Kirby Delay Discounting Task	- Trend for decreased craving (OCCS) and increased willingness to change drug use after 15 sessions of active tDCS, although not maintained at 1 month follow-up - No tDCS effects in other clinical measures of addiction	Impulsivity indicated by k value in the Kirby Delay Discounting Task improved over time
Gorini et al. (2014)	Dependent cocaine users DSM-4 criteria	tDCS	Bilateral DLPFC F3 anode/cathode F4 cathode/anode	1.5 mA 20 min Single session	Cognitive: BART, GDT	N/A	- Decreased risk taking behavior in BART after active treatment but not sham - Increase/decrease of safe bets in GDT after RAn+/LAn+ stimulation in CU patients - Increase of safe bets in GDT after RAn+ stimulation in healthy controls
Verveer et al. (2020)	Cocaine use disorder DSM-5 crieria	tDCS	Bilateral DLPFC Right anode Left cathode	2 mA 13 min 10 sessions	Addiction: Craving (Ecological Momentary Assessments), Relapse (3 month follow-up) Cognitive: 2-Choice Gambling Task. Go/No-Go*	- No significant tDCS effects on craving nor relapse - Reduction in relapse rates for crack cocaine use after active tDCS compared to sham	No significant tDCS effects on cognitive/behavioral measures

Opioid							
Aksu et al. (2024)	Opioid use disorder DSM-5 criteria	tDCS, Cognitive Training (CT)	Bilateral DLPFC Right anode Left cathode	2 mA 20 min Single session	Cognitive: BART, IGT*, SST*, Stroop*	N/A	Improved decision making under ambiguity (IGT) after active tDCS compared to sham
Borzooee et al. (2024)	Opioid use disorder under treatment DSM-5 criteria	tDCS	Right DLPFC F4 anode FP1 cathode	2 mA 13 min 10 sessions	Addiction: Instant craving (DDQ), Periodic drug craving (OCDUS), Cue-induced craving (VAS) Cognitive: BART, Go/No-Go* Other: Serum BDNF	- Periodic drug craving (OCDUS) sigfniciantly decreased in active tDCS at 2 month follow-up - No significant intervention effects on instant (DDQ) or cue-induced craving (VAS)	- Risk taking (BART)decreased in active tDCS group after stimulation and at 2 month follow-up. - Response inhibition (GNG) decreased in active group after stimulation and at 2 month follow-up. - The above effects were not significant in the gerenalized eestimating model.

Table 4. Summary of Studies on Reward Processing and Learning (n=8)

Study	Sample	Intervention Protocol			Outcome Measures (Addiction / Cognitive / Other)	Key Findings	
		Method	NIBS Target	Parameters		Clinical	Behavioral/Cognitive
Biernacki et al. (2020)	Problematic substance users (ASSIST GCR >16)	10-Hz rTMS	Left DLPFC (F3)	110% rMT 1500 pulses Single session	Cognitive: Virtual T-maze task, PST Other: Reward positivity (ERP)	N/A	- Active rTMS enhanced aMCC reward positivity (blocks 3–4; T-maze) - Trend-level improvement in reward-based approach learning - Reward positivity amplitude and approach accuracy showed a marginal correlation at block 4
McKim et al. (2021)	SUD history (DSM-5 criteria) + Healthy controls	10-Hz tACS	Bilateral DLPFC (F3/F4, Cz ref)	2 mA peak-to-peak 30 min 3 sessions	Cognitive: HABIT task (goal-directed vs habitual)	N/A	- HC: tACS increased perseverative errors - SUD: no main effect; longer addiction duration predicted greater error reduction
Perini et al. (2020)	Treatment-seeking alcohol-dependent patients (DSM-IV)	Deep rTMS (10 Hz)	Bilateral Insula (H8 coil)	120% MT 1500 pulses/session 15 sessions	Addiction: Craving (AUQ, PACS), Alcohol consumption (Peth, TLFB) Cognitive: MID task Other: fMRI (resting-state, task-based)	Craving & drinking decreased in both groups (all $p < .01$); no rTMS–sham difference (all group × time interactions $p > .3$)	No significant group nor group × stimuli interaction effects were observed in MID reward anticipation
Biernacki et al. (2025)	Problematic opioid use (≥1 year of weekly use or ASSIST >12)	10-Hz rTMS	Left DLPFC (F3)	110% rMT 2000 pulses Single session	Cognitive: Virtual T-maze task Other: Reward positivity (ERP; N200, P300)	N/A	- Significant TMS × group interaction: TMS normalized blunted reward positivity in OUD (OU active ≈ HC; OU sham < both) - TMS also increased N200 in OUD compared to sham, while no effect on P300
Song Y et al. (2025)	Internet Gaming Disorder (DSM-5, ≥6 criteria)	tDCS (cue-synchronized)	Central Parietal (Pz cathode)	1.5 mA 20 min 2 sessions	Addiction: Craving (QGU-B, VAS) Cognitive: MID task, IGT*, SST* Other: EEG (P300, alpha) during cue reactivity task	Craving and gaming time decreased on active tDCS group (QGU-B: p<.001, weeks 1-3; VAS: p<.001, weeks 2-3; gaming time: p<.001, weeks 3-4)	- No significant change in MID (d' or reaction time) - No significant tDCS effect on SST - IGT net score increased after active tDCS compared to sham at 1 week, 4 week follow-up. SST: no significant group x time interaction for all measures
Wang et al. (2022)	Methamphetamine use disorder DSM-5 criteria	HF-rTMS	Left DLPFC F3	10 Hz 100% MT 12 sessions	Addiction: Craving (VAS) Cognitive: IGT (5 blocks)	Reductions in craving following HF-rTMS but not sham Strong correlation between impulsivity and craving	Significant group differences in blocks 4 and 5 favoring HF-rTMS group
Soyata et al. (2019)	Gambling disorder DSM-5 criteria	tDCS	Bilateral DLPFC F4 anode F3 cathode	2 mA 20 min 3 sessions	Cognitive: Iowa Gambling Task	N/A	- Active tDCS increased IGT net scores (large effect size)
Aksu et al. (2024)	Opioid use disorder DSM-5 criteria	tDCS, Cognitive Training (CT)	Bilateral DLPFC Right anode Left cathode	2 mA 20 min Single session	Cognitive: IGT, BART*, SST*, Stroop* -	N/A	Improved decision making under ambiguity (IGT) after active tDCS compared to sham

Abbreviations (Table 1~4): DLPFC = Dorsolateral Prefrontal Cortex; rTMS = repetitive Transcranial Magnetic Stimulation; tACS = transcranial Alternating Current Stimulation; tDCS = transcranial Direct Current Stimulation; rMT = resting Motor Threshold; MT = Motor Threshold; PST = Probabilistic Selection Task; HABIT = Hidden Association Between Images Task; MID = Monetary Incentive Delay; IGT = Iowa Gambling Task; SST = Stop Signal Task; ERP = Event-Related Potential; SUD = Substance Use Disorder; OU = Opioid Users; HC = Healthy Controls; AUQ = Acute Urge Questionnaire; PACS = Penn Alcohol Craving Scale; QGU-B = Questionnaire of Gaming Urges-Brief; VAS = Visual Analog Scale, DDQ, OCDUS,IGT, BART = Balloon Analog Risk Taking task,, SST = Stop Signal Task, SSRT = Stop Signal Reaction Time, OCDS = Obsessive Compulsive Drinking Scale, OSPAN =Automated Operation Span, EEG = Electroencephalography, ERP = Event-Related Potentials, ACQ-SF-R = Alcohol Craving Questionnaire, rsfMRI = Resting-State Functional Magnetic Resonance Imaging, SSD = Stop Signal Delay, STRAP-R = Sensitivity to Reinforcement for Addictive and other Primary Rewards, DDQ = Desires for Drug Questionnaire, ARCI = Addiction Research Center Inventory, OCCS = Obsessive-Compulsive Cocaine Scale, AAAQ = Alcohol Approach and Avoidance Questionnaire / * Indicate tasks in each table that are unrelated to the cognitive function it focuses on.

Appendix 1. Boolean operators listed below were used in the search.

Addiction	"Substance-Related Disorders" OR "Substance Use Disorder*" OR addiction OR "Addictive Disorder*" OR SUD OR alcohol* OR “hazardous drink*” OR nicotine OR tobacco OR smoking OR cocaine OR opioid* OR methamphetamine OR stimulant OR cannabis OR marijuana OR heroin OR benzodiazepine OR gambling OR "Internet Gaming Disorder" OR "Food Addiction" OR "binge eating" OR "behavioral addiction" OR "substance misuse" OR "substance dependence"
NIBS	"Transcranial Magnetic Stimulation” OR TMS OR "theta-burst stimulation" OR iTBS OR cTBS OR "Transcranial Direct Current Stimulation"[Mesh] OR tDCS OR "transcranial alternating current stimulation" OR tACS OR "noninvasive brain stimulation" OR "non-invasive brain stimulation" OR NIBS OR neuromodulation OR neurostimulation)
Cognition	"Computational Psychiatry" OR "Models, Psychological" OR "computational model*" OR "cognitive model*" OR "cognitive task*" OR “cognition” OR “cognitive function” OR "Reinforcement, Psychology"[Mesh] OR "reinforcement learning" OR "prediction error" OR RPE OR "learning rate" OR "model-based" OR "model free" OR "Decision Making"[Mesh] OR "decision making" OR "decision-making" OR valuation OR "action selection" OR reward OR punishment OR impulsivity OR inhibition OR "delay discounting" OR "loss aversion" OR "executive function*" OR “task”

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